

Lewis's analysis of modality
Handout for *Time and Modality* seminar
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1 Lewis's analysis (review)

In the special case where ϕ is built up entirely from well-behaved (qualitative) predicates, truth-functional operators, and first-order quantifiers:

$$\begin{aligned}\textit{Restriction} \quad \Diamond\phi &\leftrightarrow_{\text{df}} \exists w(Ww \wedge [\phi]^w) \\ \Box\phi &\leftrightarrow_{\text{df}} \forall w(Ww \rightarrow [\phi]^w)\end{aligned}$$

where w is a variable not in ϕ , ' Ww ' means ' w is a Lewis-world', and $[\phi]^w$ is the result of restricting every quantifier in ϕ to things 'in w '.

2 Three puzzles

- (1) It is possible that some swans are blue $\leftrightarrow_{\text{df}}$ there is a Lewis-world such that some swans in it are blue.

First puzzle: It seems obvious that there could *contingently* be no blue swans: possibly (no swans are blue and possibly some swans are blue). But by (1), this is true iff possibly (no swans are blue and there is a Lewis-world such that some swans in it are blue), which is inconsistent.

- (2) It is possible that no swans are blue $\leftrightarrow_{\text{df}}$ there is a Lewis-world such that no swans in it are blue.
(1') Necessarily: it is possible that some swans are blue iff there is a Lewis-world such that some swans in it are blue.

Second puzzle: The S5-inconsistent 'It is possible that some swans are blue, and it is possible that it is not possible that some swans are blue' follows from (1'), (2), and the claim that some Lewis-worlds do, and some don't, contain blue swans; and (1') follows from (1) (doesn't it?).

- (3) It is possible that there are two things that are not analogically spatiotemporally related $\leftrightarrow_{\text{df}}$ there is a Lewis-world such that there are two things in it that are not analogically spatiotemporally related.

Third puzzle: The T-inconsistent 'At least two things are Lewis-worlds, and it is not possible that at least two things are Lewis-worlds' follows from (3), the claim that there are at least two Lewis-worlds, the definition of 'Lewis-world', and classical mereology.

3 An obvious solution: contextually restricted quantification

In instances of *Restriction*, the quantifiers on the left are restricted; those on the right are unrestricted.

This prompts two questions:

- (i) What is the restriction?

- (ii) How do we analyse sentences in which *unrestricted* quantifiers are embedded under modal operators?

EXISTENTIAL REDUNDANCY When F is well-behaved, $\Diamond \exists_u x Fx$ is equivalent to $\exists_u x Fx$, and $\Box \forall_u x Fx$ is equivalent to $\forall_u x Fx$. [Note: the chapter draft screwed this up.]

GENERALIZED REDUNDANCY Whenever ϕ is a closed sentence built up from well-behaved predicates, unrestricted quantifiers, and truth-functional connectives, $\Diamond \phi$ and $\Box \phi$ are both equivalent to ϕ .

4 Modally variable quantification as *de re* restricted quantification

MODAL REDUNDANCY When ϕ is a closed sentence consisting entirely of qualitative vocabulary, $\Box \phi$ and $\Diamond \phi$ are equivalent to ϕ .

‘All contingency is *de re* contingency.’

This claim naturally goes along with what we last week called ‘Basic Modal Counterpart Theory’:

BMCT

$$\Diamond \phi(\alpha) \leftrightarrow_{\text{df}} \exists x (Cx\alpha \wedge \phi(x))$$

$$\Box \phi(\alpha) \leftrightarrow_{\text{df}} \forall x (Cx\alpha \rightarrow \phi(x))$$

If we want to combine *Restriction* with MODAL REDUNDANCY, we need to say what the relevant *de re* restriction is. Most plausible answer: it’s to things in *Cosmo* (our Lewis-world). But there is no reason for Lewis to think that this is a common kind of quantifier-restriction.

How to combine this interpretation of *Restriction* with BMCT: say that the relevant counterpart relation is one on which the counterparts of any Lewis-world are all and only the Lewis-worlds.

Temporal analogue: all change is *de re* change.

5 Alternative accounts of modally variable quantification

Suppose we want to uphold *Restriction* interpreted as applying to some form of restricted quantification that is not *de re* (indicate using subscript ‘r’). Suppose for concreteness that no_r swans are blue, but some_u swans are blue, and indeed Twin, someone generally like me in a different Lewis-world, at one point comes face to face with a blue swan and uttering the words ‘Some swans are blue’ in great surprise.

- (4) No_r swans are blue.
- (5) If no_r swans are blue, the proposition that some_r swans are blue is false.
- (6) When Twin says ‘Some swans are blue’, he doesn’t say anything false.
- (7) When Twin says ‘Some swans are blue’, he doesn’t say that some_r swans are blue.

Moral: the relevant form of restriction is not commonly practiced outside of this Lewis-world. This is very puzzling unless it is a *de re* restriction!

Those who want to resist (7) may invoke the idea that it makes sense to ask whether a proposition is true or false ‘at’ a Lewis-world, and then offer (6’) as a true substitute for the false (6):

- (6’) When Twin says ‘Some swans are blue’, he doesn’t say anything false *at his Lewis-world*.

But how does that do anything to undermine the seeming obviousness of (6)?

- The ‘relativist’ diagnosis of ‘correct but false belief’ is far less tempting in this case than in the case of, e.g., people who call things ‘tasty’ that we do not think are tasty.

6 What was Lewis’s view?

Some relevant quotes from OPW:

- (A) Our *contingent* knowledge that there are donkeys at *our* world requires causal acquaintance with the donkeys, or at least with what produces them. Our *necessary* knowledge that there are donkeys at *some* worlds—even talking donkeys, donkeys with dragons as worldmates, and what have you—does not require causal acquaintance either with the donkeys or with what produces them. (OPW 112)
- (B) As other worlds are alternative possibilities for an entire world, so the parts of other worlds are alternative possibilities for lesser individuals. . . . As quantification over possible worlds is commonly restricted by accessibility relations, so quantification over possible individuals is commonly restricted by counterpart relations. (OPW 8)
- (C) But I say that *any* possible individual is a possibility, and not all possible individuals are possible worlds. Only the biggest ones are. . . . [A possible world] is a way that an entire world might possibly be. But lesser possible individuals, inhabitants of worlds, proper parts of worlds, are possibilities too. They are ways that something less than an entire world might possibly be. (OPW 230)
- (D) We wanted to ask whether two worlds could differ in their laws without differing in their distribution of local qualitative character. But if we read the ‘could’ as a diamond, the thesis in question turns into this: it is not the case that, possibly, two worlds differ in their laws without differing in their distribution of local qualitative character. In other words: there is no world wherein two worlds differ in their laws without differing in their distribution of local qualitative character. That’s trivial—there is no world wherein two worlds do anything. At any one world W, there is only the one single world W. The sentential modal operator disastrously restricts the quantification over worlds that lies within its scope. (OPW 16)
- (E) I do not suppose that [restrictive modifiers] must restrict all quantifiers in their scope, without exception. ‘In Australia, there is a yacht faster than any other’ would mean less than it does if the modifier restricted both quantifiers rather than just the first. . . . [A] lot is left up to the pragmatic rule that what is said should be interpreted so as to be sensible. (OPW 6).